CHAPTER 1

# Number Sequences

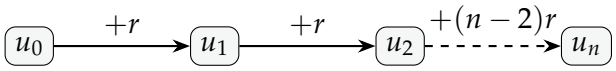
**Skills:** investigate, model, represent, calculate, communicate

## 1.1 Core Course Notes

A numerical sequence assigns to each natural number $n$ a number $u_n$. The index $n$ denotes a position in the sequence; it must not be confused with the value $u_n$.

### 1.1.1 Arithmetic sequence

**Key point.** A sequence $(u_n)$ is **arithmetic** when there is a real number $r$ such that, for all $n$, $u_{n+1} = u_n + r$. The number $r$ is the common difference. If the first term is $u_0$, then $u_n = u_0 + nr$. If the first term is $u_1$, then $u_n = u_1 + (n-1)r$.




Three numbers $x, y, z$, written in this order, are three consecutive terms of an arithmetic sequence if $y - x = z - y$, that is, if $y = \dfrac{x+z}{2}$. This last quantity is the arithmetic mean of $x$ and $z$.

**Method.** To model an additive situation: 1) define $u_n$ and state the unit; 2) identify the first term; 3) express the constant increase or decrease as the common difference $r$; 4) write the general term; 5) answer with a sentence in context.

**Example 1.** A shop sells 480 units in the first month and then 25 more units each month. With $u_1 = 480$ and $r = 25$, we get $u_n = 480 + 25(n-1)$. Thus $u_{12} = 755$: the shop expects to sell 755 units in the twelfth month.

### 1.1.2 Sum of arithmetic terms

For the $n+1$ terms from $u_0$ to $u_n$,

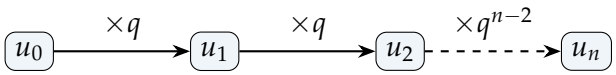
$$u_0 + u_1 + \cdots + u_n = (n+1)\frac{u_0 + u_n}{2}.$$

The formula is easy to remember as: *number of terms multiplied by the mean of the first and last terms.*

**Watch out.** In $u_3 + u_4 + \cdots + u_{12}$, there are $12 - 3 + 1 = 10$ terms, not 9 or 12.

### 1.1.3 Geometric sequence

**Key point.** A sequence $(u_n)$ is **geometric** when there is a real number $q$ such that $u_{n+1} = qu_n$. If the first term is $u_0$, then $u_n = u_0 q^n$. If the first term is $u_1$, then $u_n = u_1 q^{n-1}$.



An increase of $t\,\%$ corresponds to $q = 1 + \dfrac{t}{100}$; a decrease of $t\,\%$ to $q = 1 - \dfrac{t}{100}$. Three positive terms $x, y, z$ are consecutive in a geometric sequence if $y^2 = xz$; $y = \sqrt{xz}$ is their geometric mean. For $q \neq 1$,

$$u_0 + u_1 + \cdots + u_n = u_0 \frac{1 - q^{n+1}}{1 - q}.$$

### 1.1.4 Algorithms: computing a sum

```
def somme_carres(n):
    s = 0
    for k in range(1, n + 1):
        s = s + k**2
    return s
```

The accumulator `s` holds the partial sum; the counter `k` runs through the integers from 1 to $n$.

## Session 1 — Arithmetic sequences: recognising and modelling

Number Sequences

15 Sep 2026 • B: 2 h – C: 1 h

**Objectives.**

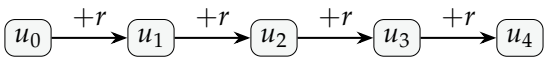
- Recognise an additive change.
- Find the common difference.
- Define a sequence in context.

**Fluency · 6 to 8 minutes, no calculator.**

1. Write the multiplier for a 10 % increase. $1 + \dfrac{10}{100} = 1.10$
2. Calculate $u_4$ if $u_0 = 14$ and $u_{n+1} = u_n + 3$. $u_4 = 26$
3. How many terms from $u_3$ to $u_{15}$? $15 - 3 + 1 = 13$
4. Give the multiplier for a 8 % decrease. $0.92$
5. Give the common difference of $12, 12 + 4, 12 + 2 \times 4, \ldots$. $r = 4$

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: recognise an additive change and find the common difference.

**Activity 1.** A club has 320 members. It expects to gain 18 more each year. Let $u_n$ be the number of members after $n$ years. Calculate $u_0, u_1, u_2$. Which operation takes you from one term to the next? Write a relation between $u_{n+1}$ and $u_n$, then predict $u_5$.




**Solution.** $u_0 = 320$, $u_1 = 338$, $u_2 = 356$; we add 18. So $u_{n+1} = u_n + 18$ and $u_5 = 410$. Have students say aloud which piece of data triggers the choice of method.

**Written record.** A sequence is arithmetic when $u_{n+1} - u_n = r$ is constant. In a real-life situation, a constant absolute increase leads to an arithmetic sequence.

**Example 2.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Solution.** $u_0 = 320$, $u_1 = 338$, $u_2 = 356$; we add 18. So $u_{n+1} = u_n + 18$ and $u_5 = 410$.

**Exercise 1.1 · essential.**

1. The sequence $(u_n)$ satisfies $u_0 = 110$ and $u_{n+1} = u_n + 3$. Write $u_n$ then calculate $u_{12}$.
2. A quantity is worth 420 and decreases by 3 % per period. Write $v_n$ and calculate $v_5$.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

**Solution.** $u_n = 110 + 3n$ and $u_{12} = 146$, for the second one, $q = 0.97$ and $v_n = 420 \times q^n$; the third item is justified by the constant difference or quotient,

**Further practice:** Calao, course notes pp. 18–21; exercises p. 24, no. 1–3. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

**Application.** A warehouse manager is testing three models. For every natural number $n$, $n$ is the week number.

- Number of parcels prepared: $a_n = 3(2n + 7) - n$.
- Number of customer complaints: $b_n = 2n^2 - n + 4$.
- Number of pallets still in stock: $c_n = 150 - 7n$.

a. For each of the three sequences, calculate $u_{n+1} - u_n$ (with $u = a$, $b$ or $c$) and state whether the sequence is arithmetic; if so, give its common difference.

b. For each arithmetic sequence, interpret the common difference in context.

**Solution.** **Target:** Recognise an arithmetic sequence by differences.

a. **Sequence $(a_n)$.** Expanding: $a_n = 6n + 21 - n = 5n + 21$. Then $a_{n+1} - a_n = 5(n + 1) + 21 - (5n + 21) = 5$. The difference is constant: $(a_n)$ is arithmetic with common difference 5 (first term $a_0 = 21$).

**Sequence $(b_n)$.** $b_{n+1} - b_n = 2(n + 1)^2 - (n + 1) + 4 - (2n^2 - n + 4) = 2(2n + 1) - 1 = 4n + 1$. This difference depends on $n$: for example $b_0 = 4$, $b_1 = 5$, $b_2 = 10$, so $b_1 - b_0 = 1$ and $b_2 - b_1 = 5$. The sequence $(b_n)$ is not arithmetic.

**Sequence $(c_n)$.** $c_{n+1} - c_n = 150 - 7(n + 1) - (150 - 7n) = -7$. The sequence $(c_n)$ is arithmetic with common difference $-7$ (first term $c_0 = 150$).

b. Each week, the warehouse prepares 5 more parcels than the week before. Each week, the stock goes down by 7 pallets (a negative common difference means a decrease).

**Homework.** Redo practice exercise 2 with the following data: replace the initial value by 125 and the rate or common difference/ratio by 3. Write it up in four lines: data, formula, calculation, conclusion.

**Solution.** The correction depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

**Method.** **Running a 1-hour session:** 7 min of fluency; 10 min of sharing and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
**Running a 2-hour session:** add 20 min of investigation, 20 min of formalising the rule, a second independent exercise and 10 min of comparative correction.
**Differentiation:** give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.

**Session 2** — **Arithmetic sequences: the general term**

Number Sequences — 17 Sep 2026 • B: 1 h – C: 2 h

**Objectives.**

- Move from the recurrence relation to the general term.
- Calculate a distant term.
- Interpret the index.

**Fluency · 6 to 8 minutes, no calculator.**

1. Simplify $3x + 5 - x + 2$. $2x + 7$
2. How many terms from $u_3$ to $u_{15}$? $15 - 3 + 1 = 13$
3. Give the multiplier for a 8 % decrease. $0.92$
4. Give the common difference of $12, 12 + 5, 12 + 2 \times 5, \ldots$. $r = 5$
5. Calculate $u_4$ if $u_0 = 15$ and $u_{n+1} = u_n + 3$. $u_4 = 27$

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: move from the recurrence relation to the general term and calculate a distant term.

**Activity 2.** A shop has 1 250 active loyalty cards at index 0 and gains 75 per month. Without calculating all the terms, find the number of cards at index $n$, then at index 18. Explain the role of 18 in the calculation.

**Solution.** $u_n = 1\,250 + 75n$, so $u_{18} = 2\,600$. The number 18 counts the increases since index 0. Have students say aloud which piece of data triggers the choice of method.

**Written record.** If $(u_n)$ is arithmetic with common difference $r$, then $u_n = u_0 + nr$ or $u_n = u_1 + (n-1)r$, depending on the index of the first term.

**Example 3.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Solution.** $u_n = 1\,250 + 75n$, so $u_{18} = 2\,600$. The number 18 counts the increases since index 0.

**Exercise 1.2 · essential.**

1. The sequence $(u_n)$ satisfies $u_0 = 120$ and $u_{n+1} = u_n + 4$. Write $u_n$ then calculate $u_{12}$.
2. A quantity is worth 440 and decreases by 4 % per period. Write $v_n$ and calculate $v_5$.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

**Solution.** $u_n = 120 + 4n$ and $u_{12} = 168$, for the second one, $q = 0.96$ and $v_n = 440 \times q^n$; the third item is justified by the constant difference or quotient,

**Further practice:** Calao, course notes pp. 18–21; exercises p. 24, no. 5–8. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

**Application.** A delivery company is gradually replacing its diesel vehicles with electric ones. At the start of 2025 (rank 0), it owns 186 diesel vehicles and it withdraws 4 of them every year. Let $d_n$ be the number of diesel vehicles at the start of the year $2025 + n$; the sequence $(d_n)$ is arithmetic.

a. Give the first term $d_0$ and the common difference of the sequence $(d_n)$.

b. Express $d_n$ in terms of $n$ for every natural number $n$.

c. Calculate $d_{12}$ and $d_{30}$. Interpret $d_{30}$ in a sentence.

**Solution.** **Target:** General term and distant terms (arithmetic).

a. $d_0 = 186$; 4 vehicles are withdrawn each year, so $d_{n+1} = d_n - 4$: the common difference is $r = -4$.

b. For an arithmetic sequence with first term $d_0$: $d_n = d_0 + nr$, that is $d_n = 186 - 4n$.

c. $d_{12} = 186 - 4 \times 12 = 186 - 48 = 138$ and $d_{30} = 186 - 4 \times 30 = 186 - 120 = 66$.
Rank 30 corresponds to the year $2025 + 30 = 2055$: according to this model, the company will still own 66 diesel vehicles at the start of 2055.

**Homework.** Redo practice exercise 3 with the following data: replace the initial value by 130 and the rate or common difference/ratio by 4. Write it up in four lines: data, formula, calculation, conclusion.

**Solution.** The correction depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

**Method.** **Running a 1-hour session:** 7 min of fluency; 10 min of sharing and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
**Running a 2-hour session:** add 20 min of investigation, 20 min of formalising the rule, a second independent exercise and 10 min of comparative correction.
**Differentiation:** give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.

## Session 3 — Sum of the terms of an arithmetic sequence

Number Sequences

22 Sep 2026 • B: 2 h – C: 1 h

**Objectives.**

- Count the terms of a sum.
- Apply the formula.
- Check the order of magnitude.

**Fluency · 6 to 8 minutes, no calculator.**

1. Calculate 20 % of 120. 24
2. Give the multiplier for a 8 % decrease. 0.92
3. Give the common difference of $12, 12+6, 12+2\times 6, \ldots$. $r=6$
4. Calculate $u_4$ if $u_0=16$ and $u_{n+1}=u_n+3$. $u_4=28$
5. How many terms from $u_3$ to $u_{15}$? $15-3+1=13$

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: count the terms of a sum and apply the formula.

**Activity 3.** A company pays a monthly bonus: 45 euros in the first month, then 5 euros more each month. Calculate the total paid over the first twelve months in three ways: explicit sum, pairing first and last terms, formula.

**Solution.** $u_{12}=45+11\times 5=100$. The total is $12\times\dfrac{45+100}{2}=870$ euros. Have students say aloud which piece of data triggers the choice of method.

**Written record.** A sum of arithmetic terms equals number of terms $\times\dfrac{\text{first}+\text{last}}{2}$. From $u_p$ to $u_q$, there are $q-p+1$ terms.

**Example 4.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Solution.** $u_{12}=45+11\times 5=100$. The total is $12\times\dfrac{45+100}{2}=870$ euros.

**Exercise 1.3 · essential.**

1. The sequence $(u_n)$ satisfies $u_0=130$ and $u_{n+1}=u_n+5$. Write $u_n$ then calculate $u_{12}$.
2. A quantity is worth 460 and decreases by 5 % per period. Write $v_n$ and calculate $v_5$.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a

difference or a quotient.

**Solution.** $u_n = 130 + 5n$ and $u_{12} = 190$, for the second one, $q = 0.95$ and $v_n = 460 \times q^n$; the third item is justified by the constant difference or quotient,

**Further practice:** Calao, course notes pp. 20–21; exercises p. 24, no. 11–13, then p. 28, no. 43–44. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

**Application.**

a. Calculate $\sum_{i=1}^{365} i = 1 + 2 + \cdots + 365$.

b. Calculate $\sum_{k=100}^{365} k = 100 + 101 + \cdots + 365$. State the number of terms in this sum.

c. To build up some savings, Lova takes on a challenge: on the $k$-th day of the year, she puts $k$ cents into a money box (1 cent on day 1, 2 cents on day 2, and so on). How much, in euros, will she have put in altogether over a 365-day year? And from day 100 to day 365 inclusive?

**Solution.** **Target:** Sum of consecutive integers.

a. The integers from 1 to 365 form an arithmetic sequence with common difference 1; there are 365 terms. So $\sum_{i=1}^{365} i = 365 \times \dfrac{1 + 365}{2} = 365 \times 183 = 66\,795$.

b. From 100 to 365 there are $365 - 100 + 1 = 266$ terms. So $\sum_{k=100}^{365} k = 266 \times \dfrac{100 + 365}{2} = 266 \times 232.5 = 61\,845$.
Check: $66\,795 - 61\,845 = 4\,950 = \dfrac{99 \times 100}{2}$, which is indeed the sum of the integers from 1 to 99.

c. The total for the year is 66 795 cents, that is €667.95. From day 100 to day 365, Lova puts in 61 845 cents, that is €618.45.

**Homework.** Redo practice exercise 2 with the following data: replace the initial value by 135 and the rate or common difference/ratio by 5. Write it up in four lines: data, formula, calculation, conclusion.

**Solution.** The correction depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

**Method.** **Running a 1-hour session:** 7 min of fluency; 10 min of sharing and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
**Running a 2-hour session:** add 20 min of investigation, 20 min of formalising the rule, a second independent exercise and 10 min of comparative correction.
**Differentiation:** give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.

## Session 4 — Geometric sequences: recognising and modelling

Number Sequences

24 Sep 2026 • B: 1 h – C: 2 h

**Objectives.**

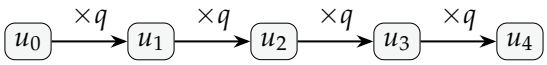
- Recognise a multiplicative change.
- Convert a rate into a multiplier.
- Compare arithmetic and geometric sequences.

**Fluency · 6 to 8 minutes, no calculator.**

1. Write the multiplier for a 25 % increase. $1 + \dfrac{25}{100} = 1.25$
2. Give the common difference of $12, 12+7, 12+2\times 7, \ldots$. $r = 7$
3. Calculate $u_4$ if $u_0 = 17$ and $u_{n+1} = u_n + 3$. $u_4 = 29$
4. How many terms from $u_3$ to $u_{15}$? $15 - 3 + 1 = 13$
5. Give the multiplier for a 8 % decrease. $0.92$

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: recognise a multiplicative change and convert a rate into a multiplier.

**Activity 4.** An online shop increases its number of visitors by 6 % per month, starting from 8 000 visitors. Calculate the first three values. Compare the differences, then the quotients. Identify the type of sequence.




**Solution.** The multiplier is 1.06: 8 000, 8 480, 8 988.8. The quotients are 1.06; the sequence is geometric. Have students say aloud which piece of data triggers the choice of method.

**Written record.** A sequence is geometric when $\dfrac{u_{n+1}}{u_n} = q$ is constant (non-zero terms). A constant relative change leads to a geometric sequence.

**Example 5.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Solution.** The multiplier is 1.06: 8 000, 8 480, 8 988.8. The quotients are 1.06; the sequence is geometric.

**Exercise 1.4 · essential.**

1. The sequence $(u_n)$ satisfies $u_0 = 140$ and $u_{n+1} = u_n + 6$. Write $u_n$ then calculate $u_{12}$.
2. A quantity is worth 480 and decreases by 6 % per period. Write $v_n$ and calculate $v_5$.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

**Solution.** $u_n = 140 + 6n$ and $u_{12} = 212$, for the second one, $q = 0.94$ and $v_n = 480 \times q^n$; the third item is justified by the constant difference or quotient,

**Further practice:** Calao, course notes pp. 23–25; exercises p. 25, no. 14–18. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

**Application.** An app is being tested before its launch. In week 0, 3 testers install it. We assume that, each week, the number of new testers is multiplied by 4. Let $v_n$ be the number of new testers in week $n$; the sequence $(v_n)$ is geometric.

a. Give the first term and the common ratio of the sequence $(v_n)$, then express $v_n$ in terms of $n$.

b. Calculate $v_5$ and $v_9$.

c. Does the model seem realistic over a long period? Justify in a sentence.

**Solution.** **Target:** General term of a geometric sequence.

a. $v_0 = 3$ and $v_{n+1} = 4v_n$: the common ratio is $q = 4$. For a geometric sequence, $v_n = v_0 q^n$, so $v_n = 3 \times 4^n$.

b. $v_5 = 3 \times 4^5 = 3 \times 1\,024 = 3\,072$ and $v_9 = 3 \times 4^9 = 3 \times 262\,144 = 786\,432$.

c. No: according to the model there would be 786 432 new testers in week 9, and over 3 million the following week ($v_{10} = 3\,145\,728$). Such fast growth cannot last: the number of potential users is limited.

**Homework.** Redo practice exercise 3 with the following data: replace the initial value by 140 and the rate or common difference/ratio by 6. Write it up in four lines: data, formula, calculation, conclusion.

**Solution.** The correction depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

**Method.** **Running a 1-hour session:** 7 min of fluency; 10 min of sharing and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
**Running a 2-hour session:** add 20 min of investigation, 20 min of formalising the rule, a second independent exercise and 10 min of comparative correction.
**Differentiation:** give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.

**Session 5** — **Geometric sequences: general term and sum**

Number Sequences — 29 Sep 2026 • B: 2 h – C: 1 h

**Objectives.**

- Write the general term.
- Calculate a geometric sum.
- Interpret a cumulative change.

**Fluency · 6 to 8 minutes, no calculator.**

1. Simplify $3x + 8 - x + 2$. $2x + 10$
2. Calculate $u_4$ if $u_0 = 18$ and $u_{n+1} = u_n + 3$. $u_4 = 30$
3. How many terms from $u_3$ to $u_{15}$? $15 - 3 + 1 = 13$
4. Give the multiplier for a 8 % decrease. $0.92$
5. Give the common difference of $12, 12 + 8, 12 + 2 \times 8, \ldots$. $r = 8$

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: write the general term and calculate a geometric sum.

**Activity 5.** A capital of 250 000 Ar is invested at 4 % per year, compound interest. Write the capital $C_n$ after $n$ years. Calculate $C_6$, then the sum $C_0 + \cdots + C_6$, and explain what this last sum represents mathematically and why it is not the final capital.

**Solution.** $C_n = 250\,000 \times 1.04^n$ and $C_6 \approx 316\,330$ Ar. The sum is the sum of seven successive values, not an actual accumulation of the investment. Have students say aloud which piece of data triggers the choice of method.

**Written record.** If $(u_n)$ is geometric with common ratio $q$, $u_n = u_0 q^n$ ou $u_n = u_1 q^{n-1}$. For $q \neq 1$, $u_0 + \cdots + u_n = u_0 \dfrac{1 - q^{n+1}}{1 - q}$.

**Example 6.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Solution.** $C_n = 250\,000 \times 1.04^n$ and $C_6 \approx 316\,330$ Ar. The sum is the sum of seven successive values, not an actual accumulation of the investment.

**Exercise 1.5 · essential.**

1. The sequence $(u_n)$ satisfies $u_0 = 150$ and $u_{n+1} = u_n + 2$. Write $u_n$ then calculate $u_{12}$.
2. A quantity is worth 500 and decreases by 2 % per period. Write $v_n$ and calculate $v_5$.
3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

**Solution.** $u_n = 150 + 2n$ and $u_{12} = 174$, for the second one, $q = 0.98$ and $v_n = 500 \times q^n$; the third item is justified by the constant difference or quotient,

**Further practice:** Calao, course notes pp. 25–28; exercises p. 25, no. 19–25, then p. 28, no. 42–44. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

**Application.** A reforestation association plants 1 500 trees in the first year of its programme. Thanks to new volunteers, the number of trees planted doubles every year. Let $u_0 = 1\,500$ be the number of trees planted in the first year and $u_n$ the number of trees planted in year $(n+1)$.

a. Justify that $(u_n)$ is a geometric sequence. Express $u_n$ in terms of $n$ and deduce $u_6$.

b. Calculate $S = u_0 + u_1 + \cdots + u_6$ and interpret this result in context.

**Solution.** **Target:** General term and geometric sum.

a. Doubling means multiplying by 2: $u_{n+1} = 2u_n$. The sequence $(u_n)$ is geometric with first term $u_0 = 1\,500$ and common ratio $q = 2$. So $u_n = 1\,500 \times 2^n$ and $u_6 = 1\,500 \times 2^6 = 1\,500 \times 64 = 96\,000$.

b. The sum has 7 terms (from $u_0$ to $u_6$) and $q \neq 1$:

$$S = u_0 \frac{1 - q^7}{1 - q} = 1\,500 \times \frac{1 - 2^7}{1 - 2} = 1\,500 \times \frac{-127}{-1} = 190\,500.$$

Over the first seven years of the programme, the association planted 190 500 trees in total.

**Homework.** Redo practice exercise 2 with the following data: replace the initial value by 145 and the rate or common difference/ratio by 7. Write it up in four lines: data, formula, calculation, conclusion.

**Solution.** The correction depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

**Method.** **Running a 1-hour session:** 7 min of fluency; 10 min of sharing and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
**Running a 2-hour session:** add 20 min of investigation, 20 min of formalising the rule, a second independent exercise and 10 min of comparative correction.

> **Differentiation:** give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.

## Session 6 — Review and formative assessment

Number Sequences

1 Oct 2026 • B: 1 h – C: 2 h

**Objectives.**

- Choose the right model.
- Use the formulas.
- Analyse one's own mistakes.

**Fluency · 6 to 8 minutes, no calculator.**

1. Calculate 5 % of 180. 9
2. How many terms from $u_3$ to $u_{15}$? $15 - 3 + 1 = 13$
3. Give the multiplier for a 8 % decrease. 0.92
4. Give the common difference of $12, 12 + 9, 12 + 2 \times 9, \ldots$. $r = 9$
5. Calculate $u_4$ if $u_0 = 19$ and $u_{n+1} = u_n + 3$. $u_4 = 31$

**Prerequisites.** Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: choose the right model and use the formulas.

**Activity 6.** Sort eight situations into three categories: arithmetic, geometric, neither. For each situation, underline the word or piece of data that justifies the choice, then choose the appropriate formula.

**Solution.** A constant absolute change indicates the arithmetic model; a constant percentage or multiplier indicates the geometric model; otherwise neither is guaranteed. Have students say aloud which piece of data triggers the choice of method.

**Written record.** Before calculating, identify: type of sequence, first index, common difference or ratio, and the term or sum asked for. The conclusion states the unit and the index.

**Example 7.** Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

**Solution.** A constant absolute change indicates the arithmetic model; a constant percentage or multiplier indicates the geometric model; otherwise neither is guaranteed.

**Exercise 1.6 · essential.**

1. The sequence $(u_n)$ satisfies $u_0 = 160$ and $u_{n+1} = u_n + 3$. Write $u_n$ then calculate $u_{12}$.
2. A quantity is worth 520 and decreases by 3 % per period. Write $v_n$ and calculate $v_5$.

3. For each sequence, say whether the model is arithmetic or geometric and justify it with a difference or a quotient.

**Solution.** $u_n = 160 + 3n$ and $u_{12} = 196$, for the second one, $q = 0.97$ and $v_n = 520 \times q^n$; the third item is justified by the constant difference or quotient,

**Further practice:** Calao, p. 29, no. 49–52. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

**Application.** For each statement, say whether it is true or false. Justify your answer.

1. The price of an annual travel pass goes up by €1.50 every year. This situation can be modelled by an arithmetic sequence with common difference 1.5.
2. A car loses 15 % of its value every year. This situation can be modelled by a geometric sequence with common ratio 1.15.
3. A sum is invested in a savings account with compound interest at an annual rate of 3 %, with no deposits or withdrawals. The capital can be modelled by a geometric sequence with common ratio 1.03.
4. A rainwater tank holds 1 000 L. Each day it loses 5 % of its contents, then 40 L are added. The volume can be modelled by a geometric sequence with common ratio 0.95.
5. The population of a town grows by 2 % per year. This situation can be modelled by an arithmetic sequence with common difference 2.
6. A loan of €2 000 is granted with simple interest at an annual rate of 6 %. The amount owed can be modelled by an arithmetic sequence with common difference 120.
7. The number of monthly visits to a website decreases by 300 every month. This situation can be modelled by a geometric sequence.

**Solution.** **Target:** Choose an arithmetic or geometric model.

1. **True.** €1.50 is added every year: $u_{n+1} = u_n + 1.5$, the absolute change is constant.
2. **False.** A 15 % decrease corresponds to the multiplier $1 - \dfrac{15}{100} = 0.85$: the sequence is geometric with common ratio 0.85, not 1.15 (which would mean an increase).
3. **True.** Each year the capital is multiplied by $1 + \dfrac{3}{100} = 1.03$: $C_{n+1} = 1.03\, C_n$.
4. **False.** We have $V_{n+1} = 0.95\, V_n + 40$: an addition comes on top of the multiplication. With $V_0 = 1\,000$: $V_1 = 990$ and $V_2 = 980.5$; $\dfrac{V_1}{V_0} = 0.99$ and $\dfrac{V_2}{V_1} \approx 0.9904$ are not equal, so the sequence is not geometric (nor is it arithmetic: $-10 \neq -9.5$).
5. **False.** A 2 % rise per year is a constant relative change: the sequence is geometric with common ratio 1.02.
6. **True.** With simple interest, the interest is calculated on the initial amount: $2\,000 \times 0.06 = 120$ euros are added each year. The amount owed therefore goes up by 120 euros per year.

7. **False.** The decrease is absolute and constant ($-300$ per month): the sequence is arithmetic with common difference $-300$.

**Homework.** Redo practice exercise 3 with the following data: replace the initial value by 150 and the rate or common difference/ratio by 2. Write it up in four lines: data, formula, calculation, conclusion.

**Solution.** The correction depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

**Exit ticket · 3 minutes.** On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

**Method.** **Running a 1-hour session:** 7 min of fluency; 10 min of sharing and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
**Running a 2-hour session:** add 20 min of investigation, 20 min of formalising the rule, a second independent exercise and 10 min of comparative correction.
**Differentiation:** give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.